

RESEARCH ARTICLE

Experimental and numerical study on performance enhancement of photovoltaic panel by controlling temperature with phase change material

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Summary

In order to enhance the electricity generation performance of photovoltaic (PV) panel, experimental and numerical studies were carried out to control the temperature by using phase change material (PCM). The PV combined with PCM (PV-PCM) device and PV-PCM with foam copper were designed, and their temperature, voltage, and output power were measured and compared with those of PV panel. The mathematical model of heat transfer for PV-PCM was established and verified by experiment. And the influences of PV-PCM tilt angle, PCM properties and thickness, and ambient temperature on the temperature control effect were studied numerically. The results illuminate that the PV panel's temperature decreases significantly with the using of PCM, which correspondingly enlarges its open circuit voltage and output power. And the PCM with foam copper has a better effect. The temperature control effect of the PV-PCM system is better with the increase of the tilt angle, the PCM thermal conductivity and thickness, and the decrease of PCM phase change temperature. But there are limited conditions for them. The higher the ambient temperature, the worse the temperature control effect. The present study has important reference value for PV performance enhancement.

KEYWORDS

electricity generation, performance enhancement, phase change material, photovoltaic, temperature control

1 | INTRODUCTION

With the decrease of fossil energy and the aggravation of environmental pollution, the demand for renewable energy is increasing. Solar energy, which is one of the most promising renewable energies, has a very good application prospects due to the advantage of clean and large amount.¹ Photovoltaic (PV) conversion is a main method of solar energy utilization.² In the past 60 years, the PV conversion efficiency ranges from 7% to 12% in the early stage³ to a maximum of about 38.8% in the

recent.⁴ Although the efficiency has increased six times, more than 60% of the solar energy turns into thermal energy and accumulates in the solar cell, resulting in the constant rise of the temperature and the reduction of the PV conversion efficiency.^{5,6} Research shows that the PV conversion efficiency will decrease by about 0.1% to 0.5% when the temperature increases by 1°C.⁷ Moreover, the aging rate of the PV panel will double for every 10°C rise in its temperature, when the temperature reaches the upper limit.⁸⁻¹⁰ Therefore, it is the key to maintaining a large PV conversion efficiency and extending the service

life that controlling the PV panel temperature within a reasonable range.

There are mainly two kinds of thermal regulation of PV panels, active approach and passive approach.¹¹ The former utilizes pumps to control the flow of water or air for cooling,¹² and the latter achieves the cooling effect without energy consumption.¹³ Controlling temperature through the phase change materials (PCM) is a passive method, which has the advantages of efficiency and low cost.¹⁴ Wang et al¹⁵ developed a novel battery thermal management system based on composite PCMs with aluminum boxes to improve the temperature uniformity of lithium batteries. Through the system temperature control test in different environments, it shows that PCM can effectively reduce the average temperature of lithium ion battery back. Sandro et al¹⁶ combined pig fat with photovoltaic panels and studied the effect of pig fat on photovoltaic panels from the perspective of PV-PCM system performance and economics. The performance of the PV-PCM system when using convective organic PCM and pork fat is compared. Müslüm et al¹⁷ established a simplified PV-PCM numerical model to analyze the working temperature of photovoltaic panels. By studying the different melting points and latent heat of PCM and other physical parameters, determine the applicable PCM in different climates.

PCM is most commonly used as an energy storage medium,¹⁸ applied in the building,¹⁹ solar collector,²⁰ heat exchanger,²¹ etc. Meanwhile, PCM is a good temperature control material because it can keep the temperature constant or fluctuate little by its large latent heat. There are many applications in electronic equipment.²² PV panel combined with PCM (PV-PCM) also has an expansive application prospect.²³ Smith et al²⁴ conducted a global analysis of the annual power generation growth of the PV-PCM system. Hasan et al²⁵ integrated paraffin with melting range from 38°C to 43°C on the PV panel to monitor its cooling effect in extremely hot environment. The performance of PV-PCM system is studied by numerical simulation, and the system economy is analyzed. Zhao et al²⁶ performed seasonal and annual analysis of the PV-PCM system.

In order to further analyze the system performance, the PV-PCM system is experimentally and numerically simulated. Stropnik et al²⁷ numerically and experimentally compared the electrical efficiency and output power of PV panels with and without PCM. Ma et al¹¹ evaluated the system structure, temperature control effect, and efficiency of PV-PCM and carried out in-depth summary on the development prospect and characteristics of PV-PCM technology. Radziemska et al²⁸ experimentally analyzed the temperature change of PV panel and PV/T/PCM. It is found that the improved PV module obtains lower temperature than the unmodified PV

module. Khanna et al²⁹ analyzed the influences of system angle, wind direction angle and speed, ambient temperature, and PCM phase change temperature on the PV-PCM system performance. Emam et al³⁰ analyzed the effect of tilt angle on the electrical and thermal efficiency of a concentrator PV system with PCM. Noura et al³¹ established a thermodynamic PV-PCM model and analyzed the effects of PCM type, thickness, wind speed, wind azimuth angle, and dust deposition density on the operation efficiency of photovoltaic cells.

Different PCMs have different effects on PV temperature control. Luo et al³² conducted numerical simulation and experimental research to control the solar panels temperature with shape-stable chain graphite composite materials. Hachema et al³³ studied the effects of PCM (white vaseline) and combined PCM on the thermal and electrical properties of PV panels. Al-Waeli et al³⁴ conducted an experimental study on the PV thermal system based on the nanoparticle combination PCM. The system temperature was maintained at 30°C during the peak period of solar radiation. The power and the electrical efficiency increased from 61 W to 121 W, and from 7.1% to 13.7%, respectively.

It can be seen from the above literature, that there are many studies on controlling the PV panel temperature with PCMs, and there are also few studies on some aspects. For example, the experimental study on the influence of PCM on the power generation performance of PV panels and the comprehensive influence of PCM amount and ambient temperature on the temperature control. The main purpose of this paper is to study the temperature control of PV panel by PCM in order to improve the efficiency of photoelectric conversion. In order to better understand the temperature control effect of PCM, experimental and numerical studies were conducted in this paper on the temperature of PV-PCM and the resulting increase in power generation performance. A PV-PCM device was designed and fabricated. The temperature and power output of PV, PV-PCM, and PV-PCM with foam copper were compared to evaluate the temperature control effect of different systems. The mathematical model of heat transfer for PV-PCM was established, and its reliability was verified by experiment. The effects of various factors on the temperature control effect were analyzed.

2 | EXPERIMENT

2.1 | Experimental setup

The experimental devices, which are listed in Table 1, mainly include PV panel, solar simulator, data logger, thermostatic bath, solar radiometer, and thermocouples.

The experimental system and its diagram are shown in Figure 1.

2.2 | Selection of PCM

PCM selection plays an important role in the passive PCM cooling and temperature control. Its thermophysical property, which affects the cooling effect significantly, is

TABLE 1 Experimental setup and instruments used in experiment

Instrument	Type
PV panel	SWM-10 W
Solar simulator	TYD-PD1
Data logger	Agilent34970A
Thermostatic bath	XODC-0515
Solar radiometer	AK-FZb
Thermocouple	K type

the first consideration. Low-cost, environmentally friendly, and other factors also need to be considered.

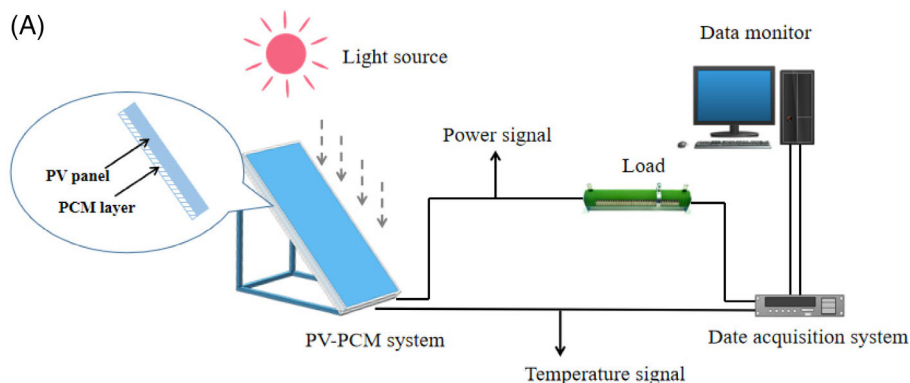
According to the above analysis, paraffin is selected as PCM in this experiment, which has good properties and widely application in many fields. The PCM thermophysical properties are shown in Table 2.

2.3 | PV panel and design of PV-PCM with and without foam copper

PV panel: The PV panel is an experimental device for photoelectric conversion, which is composed of semiconductor material, toughened glass, ethylene vinyl acetate copolymer (EVA), backplane, aluminum alloy structure, junction box, etc.

Figure 2 shows the PV panel and Table 3 lists its parameters.

PV-PCM device: A rectangular cavity formed by the copper strip and aluminum alloy frame is installed on the back of photovoltaic panel. A 9.6 mm thick rectangular cavity composed of a copper strip and aluminum alloy frame is installed on the back of the photovoltaic panel.



Schematic diagram of experimental system



Experimental system

FIGURE 1 Experimental process and system diagram, A, Schematic diagram of experimental system and B, experimental system [Colour figure can be viewed at wileyonlinelibrary.com]

TABLE 2 Thermophysical parameters of PCM

PCM	ρ (kg/m ³) (solid /liquid phase)	C_p (J/kg·K)	λ (W/m·K)	h (J/kg)	T_m (K)
Paraffin	880/780	2747	0.2	160 807	323-325

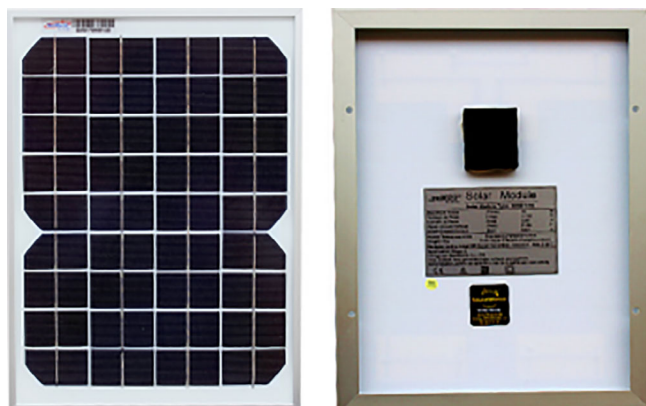


FIGURE 2 PV panel used in the experiment [Colour figure can be viewed at wileyonlinelibrary.com]

TABLE 3 PV panel parameters

PV panel	Parameter
Dimensions	290 × 240 × 18 mm
Type of batteries	156 × 156 Monocrystalline silicon
Number of batteries	36 (4 × 9)
Junction box	PVBOX17
Peak power	10 W
Peak voltage	18.17 V
Peak current	0.55 A
Open-circuit voltage	21.6 V
Short-circuit current	0.58 A

And the melted PCM is injected into the cavity to form the PV-PCM panel. Toughened glass is used as the cover plate to observe the melting of PCM. In order to avoid the leakage of liquid PCM, a rubber sealing ring is arranged between the PV panel and the toughened glass cover plate and fixed with bolts. Two holes are opened in the frame of the upper part of the PV-PCM. One is to facilitate the lead out of the PV panel wires and thermocouple wires. The other is to avoid the damage of the device due to the volume expansion when the PCM melts. Figure 3 shows the finished PV-PCM panel.

PV-PCM with foam copper: There are many ways to enhance the heat transfer of phase change materials, such as fins, heat pipes, nanomaterials, or metal foam in PCM. As a new type of multifunctional material, foamed

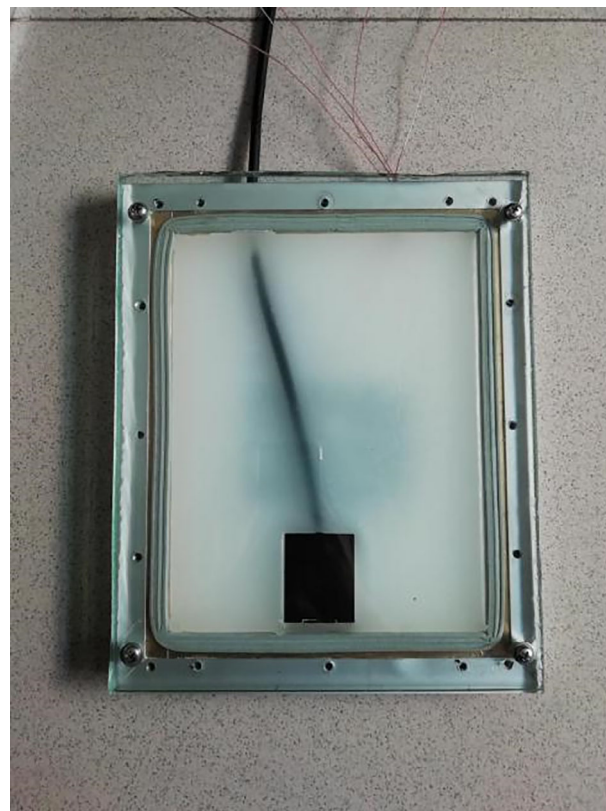


FIGURE 3 PV-PCM panel [Colour figure can be viewed at wileyonlinelibrary.com]

copper is composed of a large number of pore structures with or without connection. It has good thermal conductivity and low cost. It has great influence on the heat transfer of PCM in molten metal. With the purpose of heat transfer enhancement, foam copper is added into PCM, as shown in Figure 4A.

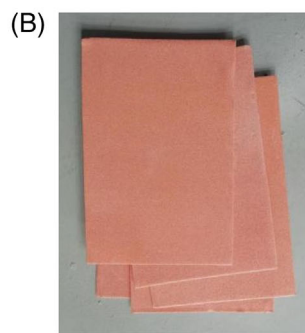
Due to the excellent conductivity performance, foam copper can be widely applied in heat dissipation of motors and electronic components. The foam copper used in this experiment is shown as Figure 4B, and its parameters are listed in Table 4.

2.4 | Temperature measure point arrangement

Ten K-type thermocouples are used to record the PV and PV-PCM panels' temperatures with the data logger



PV-PCM with foam copper in experiment



Foam copper

FIGURE 4 PV-PCM panel with foam copper. A, PV-PCM with foam copper in experiment. B, Foam copper [Colour figure can be viewed at wileyonlinelibrary.com]

TABLE 4 Foam copper parameters

Foam copper	Parameters
Specification and model	300 × 200 × 2 mm
Aperture	0.5 mm
Porosity	50%–98%
Through hole rate	≥98%
Bulk density	0.1–0.8 g/cm ³

instrument. Among them, five thermocouples (1#–5#, as shown in Figure 5) are arranged on the PV panel, and the other five thermocouples (6#–10#, as shown in Figure 5) are arranged on the PV-PCM panel. All of them are fixed with thermal conductive silicone. In addition, five thermocouples are also used to monitor the ambient temperature.

PV, PV-PCM, and ambient temperature are taken according to the average value of temperature measured by every five thermocouples arranged on them, respectively. In this way, the random error can be reduced to a small extent.

2.5 | Experimental procedures

Experimental procedures are as follows.

1. Place the PV and PV-PCM devices on the angle adjustment bracket and connect the thermocouple and resistance signal wires to the data logger in turn.
2. Turn on the artificial solar simulator and adjust the radiation intensity to 1000 W/m² according to the solar radiometer display.
3. Measure the changes of PV panel, PV-PCM, and ambient temperature over time.
4. Connect the circuit that is not connected to the load to the data acquisition instrument and measure the open circuit voltage of the photovoltaic panel and PV-PCM.
5. Connect the load resistance to the output circuit of the photovoltaic panel and PV-PCM, measure and record their output current, and calculate the output power according to the formula $P=I^2R$.
6. Observe the 1# to 10# thermocouple temperature monitored by the data collector. When the temperature of the two devices is the same or the temperature of the PV panel and the PV-PCM temperature cross, wait for the paraffin in the PV-PCM system to completely melt, the experimental test can be over.
7. Remove the glass plate covering the PCM and place the foamed copper and molten paraffin used in the experiment in the aluminum box behind the PV plate. Repeat the first six steps for the PV-PCM with foam copper device experiment.

2.6 | Error analysis

In this experiment, the uncertainty of temperature measurement mainly depends on the thermocouple. After comparing with the standard thermometer, the measurement accuracy of the thermocouple in the range of 20°C to 85°C is $\pm 0.1^\circ\text{C}$. The accuracy of electric resistance is 0.02%. And the electric current is measured by data acquisition instrument, with the accuracy of 0.01%. The accuracy of solar radiometer is 5%. The details are listed in Table 5.

3 | NUMERICAL SIMULATION

3.1 | Physical model

The PV panel used in the experiment is as the simulation object. From the light surface to the back shade surface

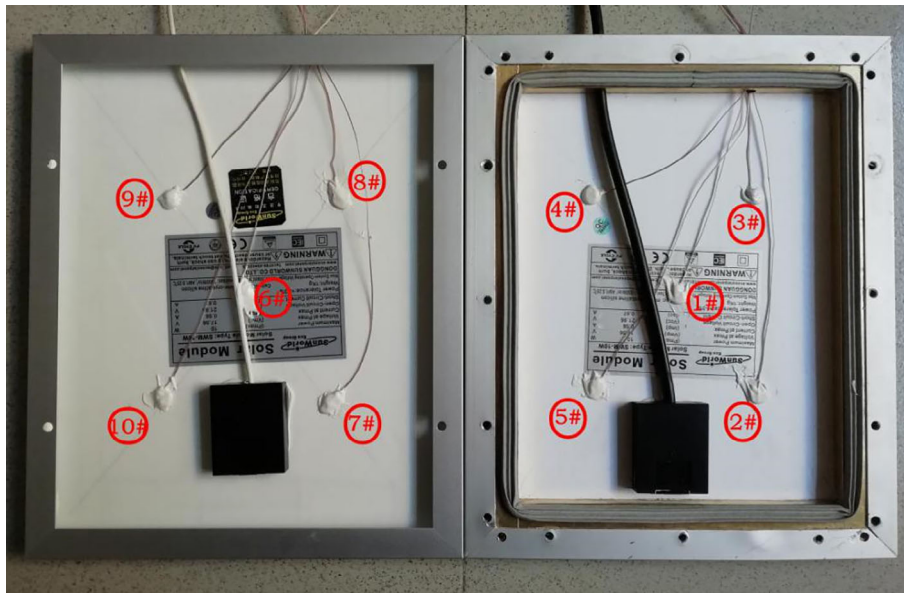


FIGURE 5 Thermocouple arrangement point [Colour figure can be viewed at wileyonlinelibrary.com]

Instrument	Model	Range	Error range
Electric resistance	ZX25a	10 mΩ-10 kΩ	0.02%
Data acquisition instrument	Agilent34970A	0–1 A	0.01%
Solar radiometer	AK-FZb	10 Wp	±5%
Thermocouple	K type	−20°C to 260°C	± 0.1°C

TABLE 5 Instrument parameters and measurement errors

of the PV panel, there are successively glass protective layer, resin sealing material EVA, crystal silicon, resin sealing material EVA, polymer layer tedlar, and aluminum plate protective layer. The PCM is inside the shell which is made of aluminum plate and tightly fixed on the PV panel. The physical property parameters of each layer are shown in Table 6.

In practice, heat, which comes from the residual solar energy after photoelectric conversion, transfers mainly along the direction of thickness, that is, from the light surface to the shaded surface (x -direction in Figure 6) for PV panel. And the heat transfers in height direction (y -direction in Figure 6) should be also considered for PV-PCM because of the convective in liquid PCM. While the heat transfer in width direction can be ignored. Based on the above analysis, the two-dimensional (2-d) numerical model is suitable for this study, as shown in Figure 6.

3.2 | Mathematical model

To simplify the study, the following assumptions are proposed.

TABLE 6 Physical property parameters of each component material of polysilicon photovoltaic panel²⁹

Material	ρ (kg/m ³)	C_p (J/kg·K)	λ (W/m·k)	δ (mm)
Glass	3000	500	1.8	3.2
EVA	960	2090	0.35	0.5
Silicon	2330	677	148	0.3
Tedlar	1200	1250	0.2	0.1
Aluminum	2675	903	211	4

1. The solar radiation on the PV panel surface is evenly distributed and treated as internal heat source with constant heat flow.
2. There are radiation and convection between the PV surface and the surrounding environment.
3. The contact thermal resistance of each part of PV-PCM module is ignored.
4. Boussinesq approximation is used to deal with the natural convection in melted PCM.

Based on the assumptions, governing equations are given as follows.

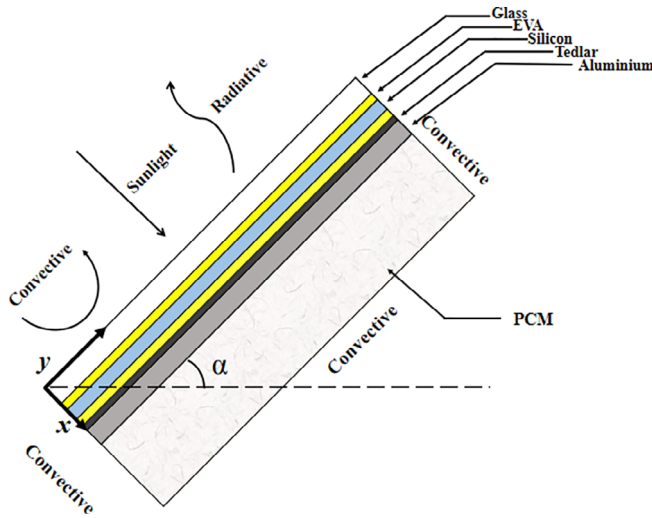


FIGURE 6 Physical model of PV-PCM system [Colour figure can be viewed at wileyonlinelibrary.com]

For PV panel (glass, EVA, silicon, tedlar, and aluminum):

$$\rho c_p \frac{\partial T}{\partial \tau} = \frac{\partial}{\partial x} \left(\lambda \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(\lambda \frac{\partial T}{\partial y} \right) + \phi, \quad (1)$$

where, ρ is the density; τ is the time; λ is the thermal conductivity; c_p is the specific heat capacity; T is the temperature; x is the term along the thickness of the PV panel; y is the term along the height of the PV panel; ϕ is the heat source in the PV panel. For EVA, $\phi = 1000 \text{ W/m}^3$. For glass, silicon, tedlar, and aluminum, $\phi = 0$.

For PCM:

Continuity equation:

$$\frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} = 0. \quad (2)$$

Momentum conservation equation:

$$\rho \left(\frac{\partial u_x}{\partial \tau} + u_x \frac{\partial u_x}{\partial x} + u_y \frac{\partial u_x}{\partial y} \right) = -\frac{\partial p}{\partial x} + \mu \left(\frac{\partial^2 u_x}{\partial x^2} + \frac{\partial^2 u_x}{\partial y^2} \right) + \rho g_x \beta (T - T_{\text{ref}}) + S_x \quad (3a)$$

$$\rho \left(\frac{\partial u_y}{\partial \tau} + u_x \frac{\partial u_y}{\partial x} + u_y \frac{\partial u_y}{\partial y} \right) = -\frac{\partial p}{\partial y} + \mu \left(\frac{\partial^2 u_y}{\partial x^2} + \frac{\partial^2 u_y}{\partial y^2} \right) + \rho g_y \beta (T - T_{\text{ref}}) + S_y. \quad (3b)$$

Energy conservation equation:

$$\frac{\partial H}{\partial \tau} + u_x \frac{\partial H}{\partial x} + u_y \frac{\partial H}{\partial y} = \lambda \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right). \quad (4)$$

In these equations, u_x is the speed of PCM in the thickness direction; u_y is the speed of PCM in the height direction; p is the pressure; μ is the dynamic viscosity; g_x is the gravity in the thickness direction; g_y is the gravity in the height direction, β is the thermal expansion coefficient. T_{ref} is the reference temperature, which equals 298 K in this study; ρ is the density of paraffin, p is the pressure, u is the velocity, H is the enthalpy and as defined by the following equation.

$$H = h + \Delta H', \quad (5)$$

$$h = h_{\text{ref}} + \int_{T_{\text{ref}}}^T c_p dT, \quad (6)$$

$$\Delta H' = f \Delta H. \quad (7)$$

In Equation (6), the reference enthalpy is represented by h_{ref} , and ΔH represents the specific melting enthalpy. Furthermore, S represents the momentum source term that can be obtained using the following equations.

$$S_x = -A(f)u_x, \quad (8)$$

$$S_y = -A(f)u_y, \quad (9)$$

$$A(f) = \frac{C(1-f)^2}{f^3 + \varepsilon}, \quad (10)$$

Where ε is set to 0.001 to make sure it is not divided by 0. C is a constant and set to 10^6 . Furthermore, f is the liquid fraction and defined as:

$$f = \begin{cases} 0 & T < T_s \\ \frac{T - T_s}{T_L - T_s} & T_s < T < T_L \\ 1 & T > T_L \end{cases} \quad (11)$$

Boundary conditions:

(a) External boundary conditions of photovoltaic panel:

For the photovoltaic panel glass surface³¹:

$$Q_{\text{glass}} = h_{g\text{-sky}}[T_{\text{PV}} - T_a] + \sigma \varepsilon_t F_{t\text{-sgy}} \left[T_{\text{PV}}^4 - T_{\text{sky}}^4 \right] + \sigma \varepsilon_t F_{tg} \left[T_{\text{PV}}^4 - T_g^4 \right]. \quad (12a)$$

Among them, the glass-atmosphere convective heat transfer coefficient ($h_{g\text{-sky}}$) is estimated by the following formula³⁵:

$$h_{g-sky} = 5.7 + 3.8V_w. \quad (12b)$$

For the side and back of the photovoltaic panel³²:

$$Q_{side \text{ and back}} = h_{w-aro}[T_w - T_a]. \quad (13a)$$

Among them, the convection heat transfer coefficient (h_{w-aro}) between the side and back of the photovoltaic panel and the surrounding air is estimated by the following formula³⁶:

$$h_{w-aro} = 2.8 + 3V_w. \quad (13b)$$

In order to facilitate the study, we take the value of V_w as the annual average wind speed of 1.7 m/s in North China, so the h_{g-sky} and h_{w-aro} are 12.16 W/(m²·K) and 7.9 W/(m²·K), respectively.

(b) Internal boundary conditions of photovoltaic panel:

In any interface of photovoltaic panel, the heat rate of the last interface is equal to that of the previous interface.

$$\lambda_{p(pv,n)} \frac{\partial T_{pv,n}}{\partial x} = \lambda_{p(pv,n+1)} \frac{\partial T_{pv,n+1}}{\partial x}. \quad (14)$$

The initial temperature of each layer of photovoltaic panel is equal to the ambient temperature.

$$T_{pv,n} = T_a. \quad (15)$$

3.3 | Numerical method

Governing equations in the present study are solved by the software, Fluent. And it is calculated by using a non-steady-state separation solver. The PISO algorithm is used for coupling the pressure field and the velocity field, the QUICK algorithm for the momentum equation and the energy equation, and the PRESTO algorithm is used to process the pressure correction equation. The convergent criterions for continuity, momentum, and energy equations are 10⁻⁶, 10⁻⁶, and 10⁻⁸, respectively. Due to relatively regular geometry of the computational domain, quadrilateral grid is used to discretize it. And the grid independent is tested by comparing the numerical results with three different grid numbers (38 357, 54 761, and 85 852) and three time steps (0.1 second, 0.2 second, and 0.3 second). The grid number of 54 761 and the time step of 0.2 second are found to be appropriate. The average surface temperature of PV panel under the three grid sets is summarized in Table 7.

3.4 | Validation of the numerical model

With the purpose of verifying the numerical method in this paper, the numerical simulation and experimental conditions are the same. And the comparison of numerical results and the corresponding experimental values is carried out. In the experiment, it is worth noting that the ambient temperature around the photovoltaic panel changes greatly, which is not consistent with the actual situation. This is because the solar radiation simulator is very large, close to the photovoltaic panel, and the experiment is carried out indoors. Therefore, as shown in Figure 7, we fit the ambient temperature curve near the photovoltaic panel according to the measured ambient temperature, and the formula is: $T = -4E - 07\tau^4 + 0.0001\tau^3 - 0.0146\tau^2 + 0.722\tau + 25.694$. From this way, this issue is taken in consider during simulation verification.

Figure 8 shows the comparison of the simulation and experimental temperature for the PV panel with the tilt angle of 45° and the irradiation intensity of 1000 W/m². It can be seen that there is good agreement of the simulation and experimental results, which proves that the numerical method used in this study is reliable.

4 | RESULTS AND DISCUSSION

4.1 | Experiment results

The temperature, the open-circuit voltage, and the output power of PV panel, PV-PCM, and PV-PCM with foam copper are obtained experimentally when the solar radiation is 1000 W/m² and the tilt angle of the system is 45°.

4.1.1 | Effect of PCM on PV panel temperature and open-circuit voltage

Figure 9 shows the temperature and open-circuit voltage of PV panel, PV-PCM system, and PV-PCM with foam copper system at solar radiation of 1000 W/m². The

TABLE 7 Results of grid-independent test for the average surface of PV panel

Grid number	Time step	T_{PV}	Difference(K)
38 358	0.2 second	339.39	-
54 761	0.2 second	337.31	2.08
85 852	0.2 second	337.26	0.05

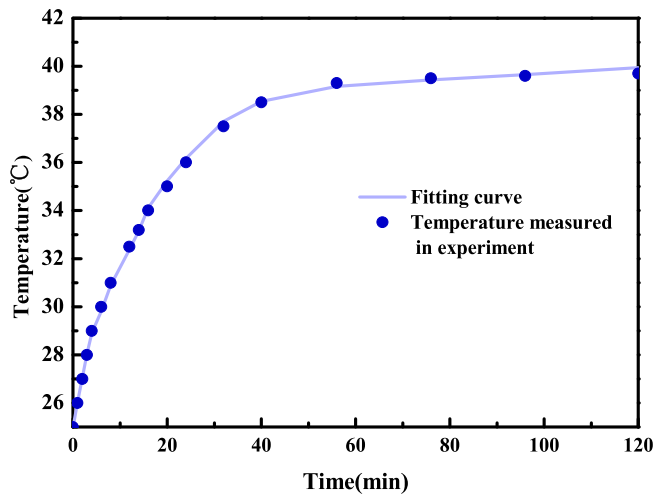


FIGURE 7 Fitting curve of ambient temperature around the PV panel [Colour figure can be viewed at wileyonlinelibrary.com]

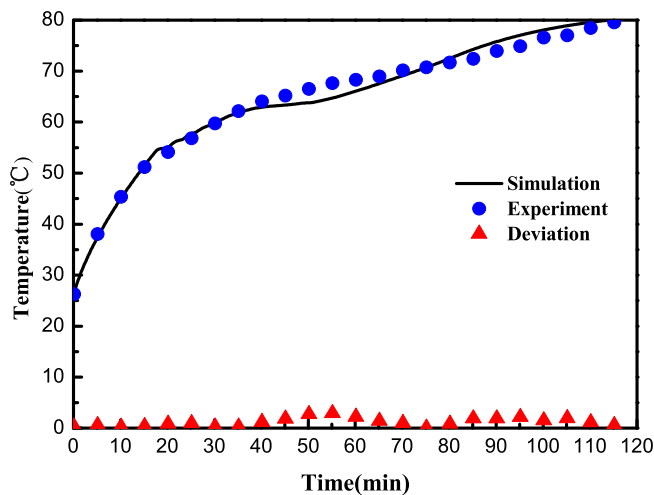


FIGURE 8 Comparison of experimental and simulation PV panel temperature [Colour figure can be viewed at wileyonlinelibrary.com]

PV panel's temperature rises rapidly in the early stage and gets to about 75°C in 20 minutes, then keeps a stable state. However, the temperature of PV-PCM goes up much more slowly. And the latter is 20°C lower than the former at 20 minutes. After that, the latter continues to rise at a slower pace, and the two temperatures are almost identical by 90 minutes. Due to the limitation of experimental equipment, the PCM thickness of PV-PCM is only 10 mm. Even though, the PV-PCM's temperature is lower than the PV panel's within 90 minutes, which means that the temperature control effect of PCM on PV panel is very obvious remarkable. This is mainly because the PCM absorbs a large amount of heat from PV panel with its great latent heat, thus keeping the operating temperature at a low level.

Figure 9 also shows the effect of the PV panel temperature on the open circuit voltage. At the beginning of experiment, the open-circuit voltages of both PV panel and PV-PCM are the maximum (24.8 V) and then decrease rapidly with the increase of temperature. That is, the relationship between open circuit voltage and temperature is inversely proportional. Within the first 20 minutes, the PV panel temperature goes up rapidly to its maximum value, and the open circuit voltage remains stable after a sharp drop from 24.8 V to 20.2 V. While, the PV-PCM temperature rises relatively slowly and its open-circuit voltage drops slowly from 24.8 V to 20.4 V in 90 minutes (the whole experimental process). During the whole experiment, the average open circuit voltage of the PV-PCM (21.36 V) is 4.2% higher than that of PV panel (20.5 V). And the former is obviously better than the latter. This shows that PCM can significantly decrease the PV panel temperature, so as to effectively improve the open-circuit voltage that is, power generation performance.

4.1.2 | Effect of PCM with copper foam on PV panel temperature and output power

Figure 10 shows the temperature and output power vs time of PV panel, PV-PCM with and without foam copper. The temperature curve of PV-PCM with foam copper is very similar to that of PV-PCM in the first 10 minutes, then the former is slightly lower than the latter. This indicates that the use of PCM combined with foam copper can further decrease (better control) the PV temperature. According to Figure 9, it can be observed that the average open-circuit voltage of PV-PCM with copper foam (22.0 V) is 7.3% and 3% higher than that of PV panels (20.5 V) and PV-PCM (21.36 V), respectively.

So far as the three output powers, we can find that they are almost the same and the maximum at the beginning, about 9 W. After that, the output power of PV drops rapidly and stabilizes at about 5.7 W after 20 minutes. That of PV-PCM drops first rapidly and then slowly to about 6 W at the end. And that of PV-PCM with foam copper has the similar trend with the second one, but the value is a little larger, about 6.1 W at the end. For quantitative analysis, the values of output power at 20 minutes are given, which are 5.87 W, 6.73 W, and 7.23 W, respectively. Compared with PV, the output powers of PV-PCM and PV-PCM with foam copper increase by 14.65% and 23.17%, respectively.

From the above analysis, PCM with foam copper can further improve the PV panel's working conditions and

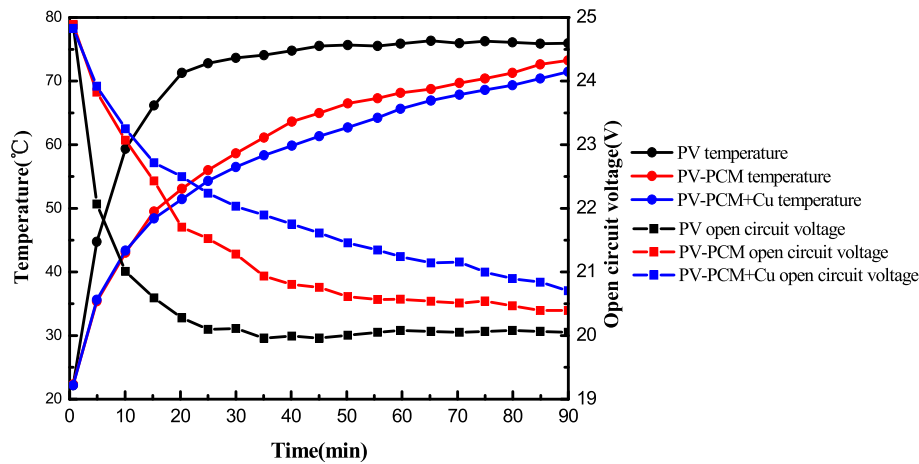


FIGURE 9 Temperature and open-circuit voltage vs time for PV panel and PV-PCM system with and without foam copper [Colour figure can be viewed at wileyonlinelibrary.com]

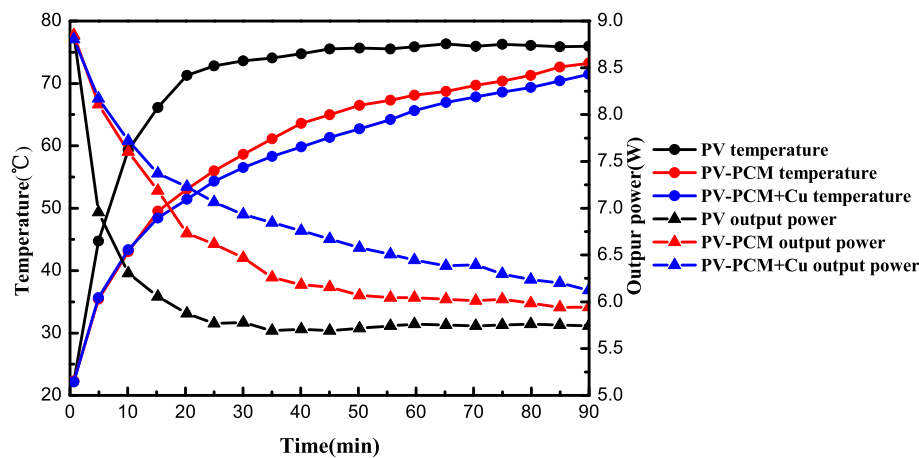


FIGURE 10 Temperature and output power vs time for PV panel and PV-PCM system with and without foam copper [Colour figure can be viewed at wileyonlinelibrary.com]

enhance its power generation performance, which, therefore, has a good application prospect.

4.2 | Simulation results

4.2.1 | Effect of the system tilt angle

The tilt angle has a great influence on the PV-PCM temperature control performance.³¹ Therefore, the temperature vs time is numerically simulated for a PV-PCM with the tilt angle of 0°, 30°, 45°, 60°, and 90°, as shown in Figure 11. At the beginning, all the temperatures for PV-PCM system at different tilt angle increase rapidly with time. And the temperature rise rates are basically the same in 30 minutes. After 30 minutes, the temperature continues rising rapidly for the PV-PCM at the tilt angle of 0°. However, it increases with a low rate at other tilt angles. By the time of 450 minutes, the temperature gets to 72°C for the PV-PCM at the tilt angle of 0°, which is much higher than those at other tilt angles, for examples, 59°C at 45° and 56°C at 90°. This is because conduction

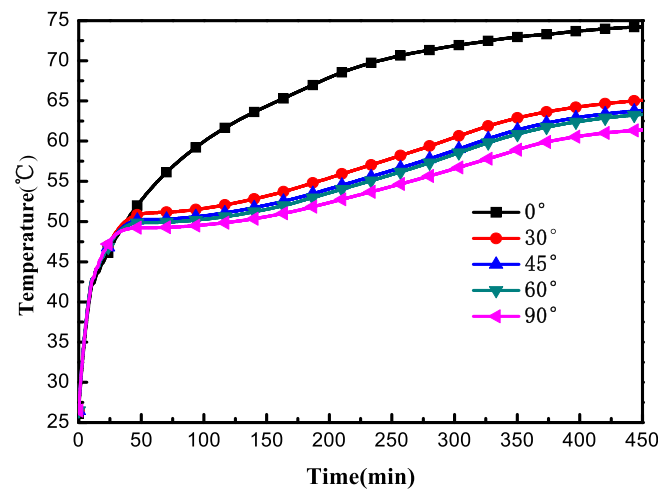


FIGURE 11 Temperature vs time for PV-PCM system at different tilt angle [Colour figure can be viewed at wileyonlinelibrary.com]

is the mode of heat transfer in the whole process when the tilt angle is 0°. While, the convection plays an important role in the tilt system, especially after 30 min.

This can be clearly seen from Figure 12, which shows the temperature, solid–liquid phase and streamline of PV-PCM at different tilt angles. According to the temperature cloud diagram in Figure 12A, it can be seen that the temperature of the photovoltaic panel reaches about 331 K at 180 minutes when the angle is 0° , and the temperature of the photovoltaic panel reaches

331 K at 450 minutes at the angle of 90° . Figure 12B shows that the greater the inclination angle, the denser the streamlines, and the stronger the convective heat transfer between PCMs.

In addition, it can also be seen from Figure 11 that the temperature decreases with the increase of tilt angle. According to Figure 12, the reason can be explained.

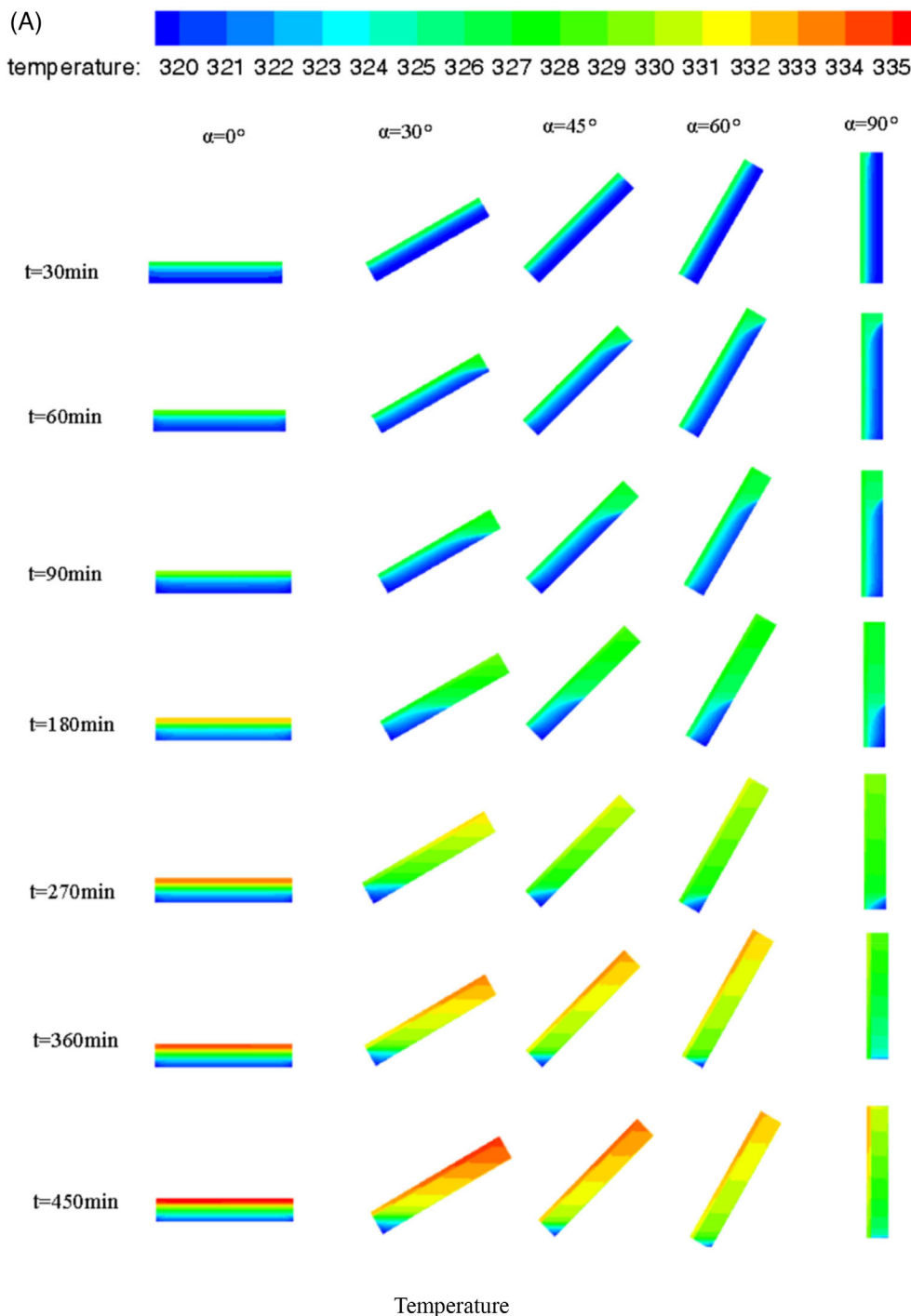


FIGURE 12 Temperature, solid–liquid and streamline vs time for PV-PCM at different tilt angles. A, Temperature. B, Solid–liquid and streamline [Colour figure can be viewed at wileyonlinelibrary.com]

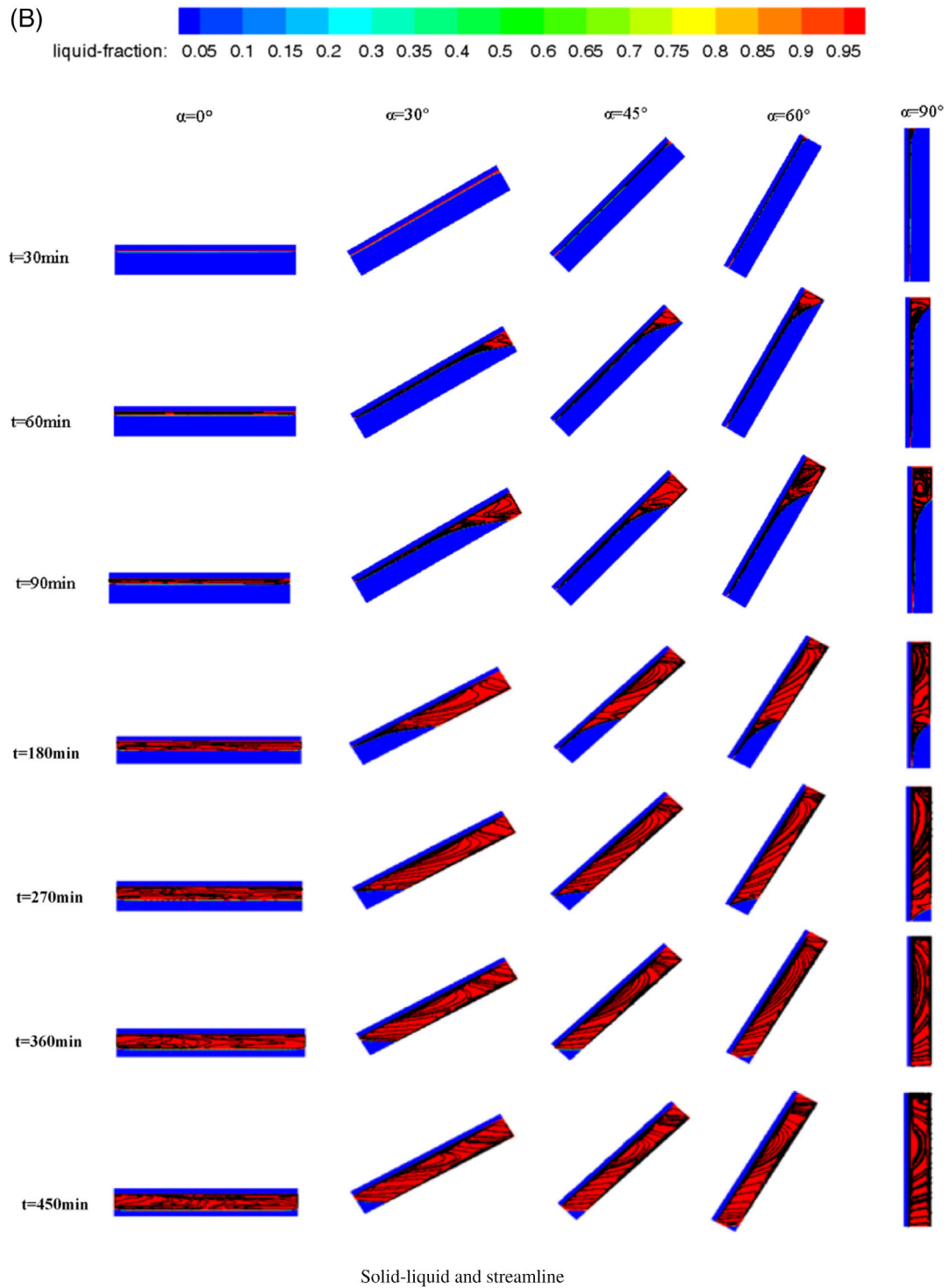


FIGURE 12 (Continued)

After 30 minutes, PCM in PV-PCM starts to melt, and the upper part melts faster than the lower part due to the natural convection. And the convection strength enhances, which make heat transfer from PV panel faster, with the increase of tilt angle.

Therefore, in order to enhance the PV performance, its tilt angle should be as large as possible in terms of temperature control. However, there is a restriction in practice, the solar radiation reception should not be weakened significantly by too large tilt angle. In addition,

the problem of uneven temperature in the height direction of PV-PCM should also be noted.

4.2.2 | Effect of PCM physical parameters

The physical parameters have a great influence on PCM melting, so the effects of the melting temperature and thermal conductivity of PCM on the PV-PCM temperature are investigated.

Figure 13 shows the temperature with the irradiation time when the melting temperature of PCM is 35°C, 37.5°C, 40°C, 42.5°C, and 45°C, respectively. As can be seen from Figure 13, the temperature of PV-PCM increased with the increase of melting temperature 270 minutes ago. While it is reversed after 270 minutes that, the higher the melting temperature, the lower the PV panel temperature. The reason for this is that all the PCMs do not melt completely before 270 minutes. And the PCM with lower melting temperature makes a lower temperature of PV-PCM. After 270 minutes, the PCM with low melting temperature gradually melts completely, then the temperature increases rapidly. In this study, 270 minutes is considered as the turning point. Generally speaking, PCM with low melting temperature has better temperature control effect on PV-PCM device. So the PCM phase transition temperature should be as low as possible. However, for PCM with low melting temperature, the PV temperature will rise after a certain period of time. This is because the PCM with low melting temperature experiences less heat loss and shorter effective time for PV thermal management. For example, close to the ambient temperature, the PCM may melt ahead of time to advance the turning point, which in turn increases the temperature of the PV panel. Therefore, PCM with low melting temperature can be improved by increasing the thickness and latent heat of phase change materials.

Figure 14 shows the temperature with irradiation time when the PCM thermal conductivity is 0.2 W/m·K, 0.4 W/m·K, 0.6 W/m·K, 0.8 W/m·K, and 1.0 W/m·K, respectively. With the thermal conductivity increasing, the temperature declines, but the decline range gradually decreases. For example, when the thermal conductivity is 0.2 W/m·K, 0.6 W/m·K, and 1 W/m·K, the PV-PCM temperature is 57.31°C, 54.90°C, and 54.13°C, respectively. The temperature difference of the second and the first is 2.41°C, and that of the third and the second is only 0.77°C. This is because PCM with high thermal conductivity transfers more heat at the same time. That is to say, it can significantly improve the temperature control performance of PV-PCM system that increasing PCM thermal conductivity, but the effect gradually weakens.

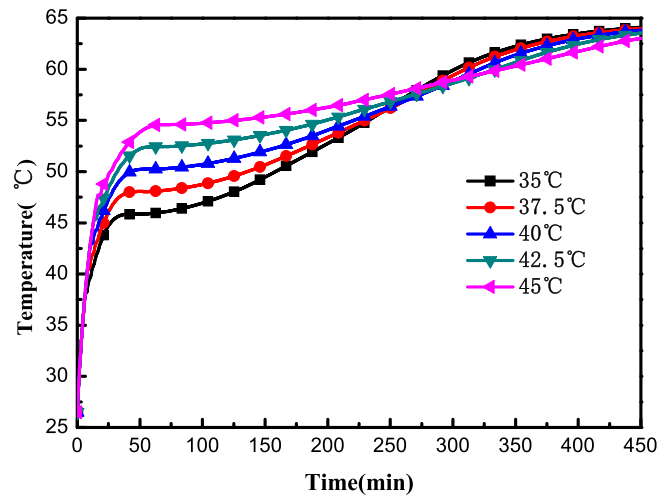


FIGURE 13 Temperature vs time of PV-PCM system with different phase change temperatures [Colour figure can be viewed at wileyonlinelibrary.com]

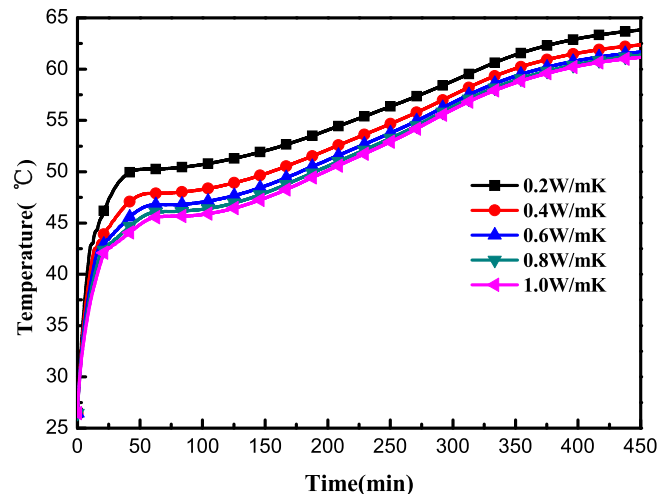


FIGURE 14 Temperature vs time for PV-PCM with different thermal conductivity [Colour figure can be viewed at wileyonlinelibrary.com]

4.2.3 | Effect of PCM thickness and ambient temperature

The amount of PCM in PV-PCM and ambient temperature are the two important factors that affect the PV-PCM temperature control performance.

Figure 15 shows the temperature of the PV-PCM system with the PCM thickness of 30 mm, 35 mm, 40 mm, 45 mm, and 50 mm, respectively. As expected, the greater the PCM thickness, the lower the PV-PCM temperature. But at the same time, this also means that the cost rises. Therefore, in practice, it is necessary to comprehensively increase the material cost and the power generation power caused by the increase in the amount of PCM to optimize the PCM thickness.

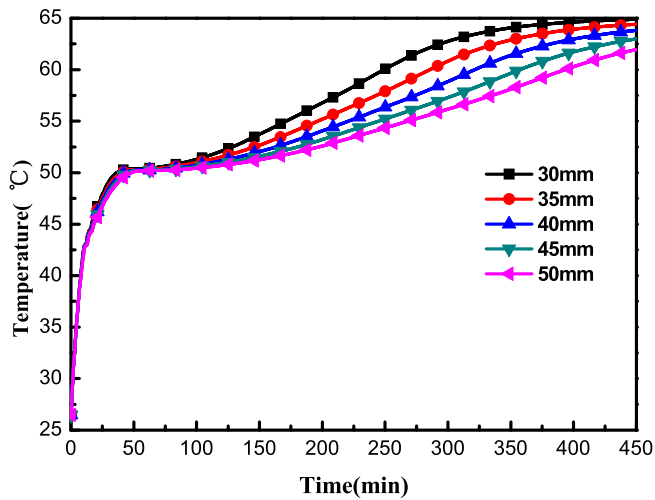


FIGURE 15 Temperature vs time for PV-PCM system with different PCM thickness [Colour figure can be viewed at wileyonlinelibrary.com]

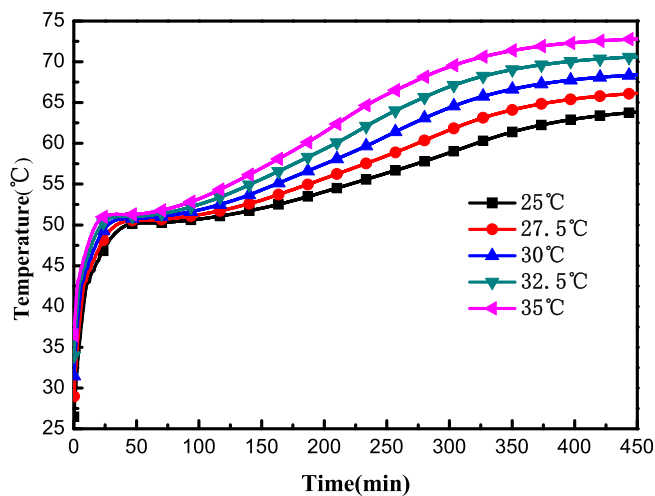


FIGURE 16 Temperature vs time of PV-PCM system of PCM under different ambient temperatures [Colour figure can be viewed at wileyonlinelibrary.com]

Figure 16 shows the PV-PCM temperature with time under different ambient temperatures. The PV-PCM temperature rises with the increase of ambient temperature, and the temperature increase range is much more high with the irradiation time. For example, when the ambient temperature goes up from 25°C to 35°C, the PV-PCM temperature rises from 52°C to 57°C at 150 minutes, and from 61°C to 72°C at 400 minutes. This is because when the ambient temperature increases, the melting time of PCM becomes shorter. After complete melting, the thermal management effect of PCM on PV decreases and the PV temperature increases. Therefore, the ambient temperature has a significant effect on the temperature control performance of the PV-PCM system. When the

ambient temperature increases, it is necessary to reduce the temperature of PV-PCM by some methods, such as increasing the thickness and thermal conductivity of PCM or adopting PCM with higher melting temperature.

5 | CONCLUSIONS

The purpose of this paper is to enhance the PV panel's performance by control its temperature with PCM. A PV-PCM device was designed and fabricated, and the temperature, open-circuit voltage, and output power of PV panel, PV-PCM, and PV-PCM with foam copper with time were experimentally measured and compared. The effects of inclination, PCM properties and thickness, and ambient temperature on the PV-PCM temperature were numerically analyzed. The conclusions show that:

1. PCM can significantly reduce the temperature of PV panel, so as to enhance the power generation performance. Adding high thermal conductivity materials such as foam copper into PCM can further improve the working conditions of the PV panel.
2. The larger the inclination angle, the better the temperature control effect of the PV-PCM device. The inclination angle should be as large as possible under the premise of good solar radiation reception.
3. The temperature control effect of PV-PCM decreases with the decrease of PCM phase transition temperature. The increased thermal conductivity can significantly improve the temperature control effect of the PV-PCM, but the effect gradually weakens.
4. The greater the thickness of PCM, the better the temperature control effect of PV-PCM, but the higher material cost. The optimal PCM thickness is obtained by weighing these two aspects.
5. When the ambient temperature rises, PV panel temperature rises, and the temperature control effect of PV-PCM becomes worse with the irradiation time. And some methods can be taken to mitigate the temperature rising, such as increasing the thickness and thermal conductivity of PCM or adopting higher melting temperature PCM.

Using PCM can control the temperature of PV panel and improve its power generation performance, but the efficiency is about 20%, and the rest of the solar energy is converted into heat energy. Therefore, in the future, the research on the utilization of waste heat of PV panels needs more in-depth research. The combination of PV, PCM, and thermoelectric materials can improve the overall power generation efficiency.

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NOMENCLATURE

Abbreviations

EVA ethylene vinyl acetate
PCM phase change material
PV photovoltaic

Symbols

c_p specific heat at constant pressure (J/kg·K)
 h latent heat (J/kg)
 T temperature (K)
 T_m melting point of PCM (K)
 u_x, u_y velocity component in the x, y coordinates (m/s)
 g gravity (m/s²)
 F view factor
 $h_{g\text{-}sky}$ convective heat transfer coefficient between glass and atmosphere (W/m²·K)
 $h_{w\text{-}aro}$ convection heat transfer coefficient (hw-aro) between the side and back of the photovoltaic panel and the surrounding air (W/m²·K)
 V_w wind velocity (m/s)
 C the parameters of the mushy region · set to 10⁶
 Q_{glass} heat loss rate of glass surface (W/m²)
 Q_{side} heat loss rate on the side and back of the photovoltaic panel (W/m²)
and back

Greek letters

ρ density (kg/m³)
 λ thermal conductivity (W/m·K)
 δ each material thickness (mm)
 σ black-body radiation constant (W/m²·K⁴)
 ε surface average emissivity
 τ time (s)
 ϕ heat source in PV panel (W/m³)
 μ dynamic viscosity (kg/m·s)
 β thermal expansion coefficient (1/K)

Subscripts

sun sunlit surface
 x PV panel thickness direction

y PV panel height direction
 a ambient air temperature
sky average temperature of the earth's surface
 w side and back of photovoltaic panel
aro around PV panel
 g ground
 t top surface
 n nth layer of the PV
ref reference
 L liquid
 S solid

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